

Rishav Saigal

Senior Data Scientist | Machine Learning Engineer | Generative AI / Agentic AI Engineer | AI Architect

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PROFESSIONAL SUMMARY

Senior Data Scientist and ML/GenAI Engineer with **8+ years** of experience architecting and deploying **enterprise-grade, production-ready AI/ML systems** across healthcare, supply chain, and media verticals. Deep expertise in **Generative AI (GenAI)**, **Agentic AI** (Google ADK, LangGraph, CrewAI), **Retrieval-Augmented Generation (RAG)**, **Prompt Engineering**, **Large Language Model (LLM) applications**, and **Time Series MLOps**. Experienced in production AI architecture, autonomous ML orchestration, multi-agent systems, and enterprise AI deployment. Author of **10 independent technical articles** (Towards AI, Towards Deep Learning, Medium, LinkedIn) plus an open, DOI-indexed **research framework & dataset** (Zenodo) on LLM behavioral drift, a self-coined **calcification effect**, multi-agent robustness, autonomous ML, and AI agent production economics. Active open-source contributor (PyCaretAgent, CognitoEDA, LLM Drift Experiment, model-router).

TECHNICAL SKILLS

- **Generative AI & LLMs:** Generative AI (GenAI), Large Language Models (LLMs), Agentic AI, Multi-Agent Systems, Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, LangGraph, CrewAI, Google ADK, Hugging Face
- **Machine Learning & Statistical Modeling:** Scikit-Learn, TensorFlow, Keras, PyCaret, SHAP / Explainable AI (XAI), Transfer Learning, Survival Analysis, Causal Inference, Time Series Forecasting
- **MLOps & Data Engineering:** MLflow, DataRobot, Docker, CI/CD for ML, Model Monitoring, Automated Retraining, Feature Engineering, PySpark, Dask, Pandas, NumPy, SciPy
- **Cloud & AI Platforms:** AWS (SageMaker, Lambda, ECS, Transcribe, Comprehend), Azure ML, GCP (Vertex AI, Cloud Run, Gemini Enterprise)
- **Programming, APIs & Databases:** Python, JavaScript, FastAPI, Flask, REST APIs, MSSQL, MySQL, PostgreSQL, Redis
- **Visualization & Applications:** Streamlit, DASH, Plotly, Seaborn, Matplotlib, Power BI, ipywidgets
- **Leadership & Architecture:** Team Leadership, Agile/Scrum, Technical Roadmapping, Stakeholder Management, Effort Estimation, Architectural Design, Proof of Concept (POC) Management, Impact & Risk Assessment, Cost Optimisation

PROFESSIONAL EXPERIENCE

Cognizant | Kolkata

Manager (Senior Data Scientist / AI Architect) | Nov 2023 – Present

- **Solution Architect & Agentic AI Lead:** Spearheaded end-to-end architectural design of enterprise-grade Generative AI and Agentic AI systems on **complex GCP infrastructure (Vertex AI, Cloud Run, Gemini Enterprise)** optimised for high-quality agent delivery at scale; drove Agile sprint planning, effort estimation, and cross-functional technical delivery; led **client deliverable management** — end-to-end ownership from requirement workshops through production handoff, stakeholder-facing POC demonstrations, and sign-off governance for high-impact GenAI initiatives.
- **Team Leadership & Architectural Direction:** Led and mentored a cross-functional team of ML/GenAI engineers; defined architectural standards, technical roadmaps, and design patterns for agentic AI initiatives; drove code reviews, sprint ceremonies, and knowledge-sharing sessions ensuring consistent delivery quality across enterprise GenAI programmes.
- **Generative AI (GenAI) Video Generation (Google ADK, LangGraph, GCP Vertex AI, Gemini Enterprise, FastAPI):** Architected a production-ready multi-agent agentic AI pipeline on **GCP/Gemini Enterprise** ingesting meeting transcripts, applying LLM-powered NLP summarisation and storyboarding, autonomously synthesising structured sales pitch videos; integrated **token-usage observability** (cost-per-run dashboards, quality-vs-cost trade-off controls) and a **self-correction loop** (automated output validation with LLM-driven retry on failure); cut content delivery from days to **~6 hours**, reducing manual effort by **30–40%**.
- **Price Elasticity Framework (Hierarchical Linear Models, MLflow, Python, Azure ML):** Engineered a dynamic pricing solution with real-time competitor data ingestion and causal inference signals; implemented CI/CD-driven automated model retraining via MLflow; projected **10–15% revenue and market share uplift**.
- **Large Language Model (LLM) Ecosystems & Retrieval-Augmented Generation (RAG) Applications (LangChain, FastAPI, Streamlit):** Built RAG-powered analytics chatbots enabling natural-language querying over enterprise databases with dynamic chart generation; engineered prompt-engineering QA toolchain auto-generating Gherkin/TMMi test cases, cutting manual testing effort significantly.

Genpact | Bengaluru

Assistant Manager | Jun 2022 – Nov 2023

- **Team Leadership & Machine Learning Architecture:** Managed and mentored a cross-functional team of 6 ML engineers; led Agile architecture reviews, sprint delivery, and end-to-end MLOps solution deployment; served as primary technical point-of-contact for executive stakeholder demos and quarterly business reviews.
- **Healthcare Consumer Analytics (Scikit-Learn, DataRobot, Azure ML, Python):** Designed and deployed production ML models to identify optimal account-level promotional strategies; applied causal inference, customer segmentation, and SHAP-based explainability (XAI); delivered **projected \$2.3M in cost savings at 90%+ model accuracy** across all product lines.
- **Stakeholder Delivery & Responsible AI:** Translated complex business requirements into scalable, production-ready ML specifications; conducted end-to-end solution demos and upskilled client teams on model interpretability, responsible AI governance, and proactive model monitoring.

Capgemini | Bengaluru

Consultant | Apr 2018 – Jun 2022

- **Technical Lead & Full-Lifecycle Delivery:** Led cross-functional teams of up to **15 members** across supply chain, manufacturing, and service analytics; owned end-to-end ML delivery lifecycle — from Agile requirement gathering through CI/CD-driven production deployment on AWS SageMaker and Azure ML infrastructure.
- **Demand Sensing (PySpark, TensorFlow, Keras, MLflow, AWS SageMaker):** Engineered scalable end-to-end Time Series MLOps pipeline with automated feature engineering, model selection, and continuous retraining; achieved **5% accuracy improvement**, **10% better model selection**, reduced training from 3 days to 2 hours, and **60% reduction** in manual intervention.
- **Predictive Maintenance (XGBoost, Cox Proportional Hazard, Python, lifelines):** Applied time-varying survival analysis and transfer-learning-enhanced XGBoost to forecast EV component failures within defined windows; achieved **82% validation accuracy** and delivered interactive Plotly/ipywidgets visualisations for operational decision-making.
- **Additional Projects:** Service Now Analytics — LSTM/MLP deep learning models for intelligent ticket volume forecasting with custom routing algorithms (**85% validation accuracy**); Computer Vision POC — transfer-learning-based brand recognition system (OpenCV, TensorFlow) for automated image-level brand detection.

EDUCATION

Indian Institute of Technology Patna — Ph.D. — Computer Science & Engineering (Pursuing)

Jul 2026 – Jun 2029 (Expected)

Birla Institute of Technology and Science, Pilani — M.Tech — Data Science & Engineering

Sep 2019 – Aug 2021

RCC Institute of Information Technology (MAKAUT) — B.Tech — Computer Science & Engineering

Aug 2013 – Jul 2017

RESEARCH INTERESTS

LLM behavioral drift, adversarial robustness, and a self-coined calcification effect in multi-agent debate systems; autonomous ML orchestration and agentic pipeline design (AutoML, self-correcting agents); probabilistic time series forecasting with LLMs; production economics and efficiency optimisation of large-scale AI agent deployments.

PUBLICATIONS & INDEPENDENT RESEARCH

The Specialized Frontier: An Inquiry into Gated AI Architectures and the Cooperative Safety Flywheel — Towards AI | Aug 2026 | [Article]

Claude Model Routing: Stop Scoring "Complexity." Score These Two Things Instead. — Towards Deep Learning | Jul 2026 | [Article]

LangGraph Multi-Agent Architecture: Building a Self-Critiquing AI Debate System — Towards AI | May 2026 | [Article]

LLM Drift Experiment: A Quantitative Framework for Measuring Behavioural Drift via Adversarial Multi-Agent Debate — Zenodo | May 2026 | [Research Dataset & Framework | DOI]

Measuring Behavioral Drift in LLMs: 22 Signals, 5 Dimensions, and the Calcification Effect — Towards AI | May 2026 | [Article]

LLM Drift Explained: Do AI Models Lose Themselves Under Adversarial Pressure? — Towards AI | May 2026 | [Article]

The \$1.5 Million Difference: Why Benchmarks are Only 10% of the AI Agent Story — LinkedIn | Mar 2026 | [Article]

AutoML on Autopilot — Towards AI | Mar 2026 | [Article]

CognitoEDA — Medium | Aug 2025 | [Article]

Beyond Prediction: Generative AI's Probabilistic Future in Time Series — Medium | Jun 2025 | [Article]

Unlocking Time Series with LLMs: A New Era with TimeCAP — Medium | Jun 2025 | [Article]

SELECTED PROJECTS

Generative AI (GenAI) Video Generation

Jan 2026 – Present

Multi-agent AI pipeline using Google ADK, LangGraph, GCP Vertex AI, and Gemini Enterprise for transcript ingestion, LLM-powered summarisation, storyboarding, automated video synthesis, token-cost observability, and self-correcting output validation.

Media Use Case (POC)

Jun 2025 – Oct 2025

Agentic AI system using CrewAI, LangGraph, and GCP for automated ad-performance analysis, multi-channel budget optimisation, and recommendation engines; projected 5–10% ROI uplift and 5–7% subscription growth.

Price Elasticity (POC)

Jun 2025 – Jul 2025

Dynamic pricing architecture utilising Hierarchical Linear Models (statsmodels, Python) with real-time competitor data ingestion and automated MLflow continuous retraining pipelines.

Enterprise Analytics Chatbot

Jul 2024 – Dec 2025

LLM-powered conversational analytics agent (LangChain, Streamlit, PostgreSQL) enabling natural-language SQL queries over enterprise databases with dynamic Plotly chart generation.

Intelligent QA / BA Chatbot

Apr 2024 – Apr 2025

Conversational AI system (LangChain, FastAPI, Python) for enterprise policy querying, action execution, and automated test-case generation in Gherkin and TMMi standards.

LLM-Driven Predictive Modelling

Jul 2024 – Sep 2024

Feature engineering pipeline converting unstructured domain text into dense numerical embeddings (OpenAI API, Scikit-Learn) coupled with a SHAP explainability layer for interpretability.

LLM-Based Test Case Generator

Nov 2023 – Mar 2024

QA automation toolchain (LangChain, FastAPI, Python) auto-generating structured Gherkin/TMMi test suites directly from unstructured business requirement documents.

Healthcare Consumer Analytics

Jun 2022 – Oct 2023

Production ML architecture (Scikit-Learn, DataRobot, Azure ML) for account-level promotional strategy optimisation, combining customer segmentation, causal inference, and SHAP explainability.

Demand Sensing MLOps

Mar 2020 – Jun 2022

End-to-end Time Series MLOps pipeline (PySpark, TensorFlow, MLflow, AWS SageMaker) featuring automated feature engineering, distributed model selection, and continuous retraining workflows.

Predictive Maintenance

Dec 2019 – Mar 2020

Time-varying survival analysis pipeline (Cox PH, XGBoost, lifelines, Python) forecasting EV component degradation windows paired with interactive Plotly operational dashboards.

Service Now Ticket Analytics

Oct 2019 – Dec 2019

Ticket volume forecasting architecture (LSTM, MLP, TensorFlow, MySQL) featuring custom routing heuristics for intelligent workload balancing and agent dispatch.

Supply Chain Demand Forecasting (POC)

Apr 2019 – Oct 2019

Demand forecasting pipeline (statsmodels, NLTK Vader, pandas, Python) integrating social media sentiment indices as external dynamic regressors with grid-search hyperparameter optimisation.

Retail Video Analytics (POC)

Apr 2018 – Mar 2019

Computer vision pipeline (OpenCV, TensorFlow, Keras, Python) performing automated brand recognition and visual asset detection across retail video streams.

CERTIFICATIONS

- **Generative AI Practitioner Certificate** — Dataiku Academy, Jun 2026. [Credential]
- **Microsoft Certified: Azure AI Fundamentals** — Microsoft, Jun 2025. [Credential]
- **MLOps Practitioner Certificate** — Dataiku Academy, Aug 2023. [Credential]
- **Advanced Designer Certificate** — Dataiku Academy, Jul 2023. [Credential]
- **ML Practitioner Certificate** — Dataiku Academy, Jul 2023. [Credential]
- **Core Designer Certificate** — Dataiku Academy, Jul 2023. [Credential]